GUIJIA ZHANG

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Trustworthy Agents · Automated Research · LLM Evaluation

Two first-author papers accepted to Findings of EMNLP 2026. Developed PaperFuture and FarField-v2 for literature-grounded ideation, experiment orchestration and reproducible verification.

EDUCATION

Shenzhen University Sep 2022 - Jun 2026

B.Eng. in Electronic Information Engineering, Elite Innovation Program | Academician Scholarship

SELECTED PUBLICATIONS

Hallucination as Exploit: Evidence-Carrying Multimodal Agents

First author | Findings of EMNLP 2026 | Paper | Synced coverage

Designed evidence-certified action authorization using constrained DOM/OCR/AX verifiers and a deterministic gate; observed 0/1,700 canonical gate bypasses after hardening across 17 attack categories.

STARS: Skill Triage via Activation Risk Scoring for LLM Agents

First author | Findings of EMNLP 2026 | Paper

Built invocation-time risk scoring and SIA-Bench with 3,000 records across seven attack families; improved high-risk AUPRC by 12% over the strongest proxy baseline on held-out prompt injection.

When Token-Level Divergence Misranks Prompt-Injection Robustness

First author | Under Review

Audited prompt-injection metrics in masked diffusion LMs; demonstrated controlled cases where lower token-level KL misranks action safety, and derived decision-aware margin conditions for authorization preservation.

Rollback as Authority Revocation: Binding Caches to Committed History

First author | Under Review

Formalized rollback as authority revocation and built a provenance-bound KV-cache adapter; identified unauthorized actions in 43/63 controlled settings across seven model families and restored authorized behavior in 36 edited tests.

When Can Historical Scores Be Reused After Judge Replacement?

First author | Under Review

Developed decision-policy criteria for reusing historical LLM-judge scores, deriving finite-pool audit guarantees and budget frontiers; analyzed certification and fallback costs across 18 saved gateway comparisons.

Do GUI Agents Believe Their Eyes? Diagnosing State-Belief Reliance on Pixels versus Structure

First author | Under Review | Paper

Separated visual perception from state belief using 735 web/mobile/desktop probes, including 225 natural divergences from 38 websites; diagnosed structural reliance despite correct visual perception across seven models.

The Commitment Window: An Operator-Relative Late Override Span under Conflicting Evidence

Second author | Under Review

Studied late-layer causal override under conflicting evidence across nine models; donor interventions reversed 61-86% of gated preferences versus at most 2.4% at matched controls.

Agent Skill Regulation as a Procedural Knowledge Representation Problem

Co-first author (second listed) | Under Review

Represented skills across syntax, semantics and execution, aligning operation and object flows to expose runtime-checkable commitments and recovery actions.

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AUTORESEARCH SYSTEMS

PaperFuture 2026

Literature-grounded research ideation | GitHub

- Developed a literature-to-idea workflow with evidence-grounded development graphs, mechanism-level proposals, multi-model conditional simulation and branch-isolated memory; retained disagreements and falsifiable experiment plans.
- Built versioned corpora across four research areas: 25,059 deduplicated paper versions and 568 full-text papers, with frozen snapshots, budget accounting, resumability and offline replay.

FarField-v2 2026

Evidence-gated automated research runtime | GitHub

- Built a state-driven runtime coordinating literature, hypotheses, competing explanations, experiments and review objections, with adaptive scheduling, redesign and replayable research state.
- Implemented preregistered two-arm probes, scientific-object binding, provenance checks and reproduction gates; separated generated proposals from empirical evidence and audited failures through offline regression and bounded live runs.

WORLD MODEL AND SELF IMPROVEMENT

Exploratory research Competence dynamics and curriculum planning

Exploring world models for verifiable code and mathematics tasks, using capability profiles as states and training curricula as actions for curriculum planning and recursive self-improvement. Studying how review constraints affect exploration diversity, data coverage and verifiable feedback.

RESEARCH AND INDUSTRY EXPERIENCE

King Abdullah University of Science and Technology Jan 2026 - Present

Remote research collaboration | Di Wang and Shu Yang

Developed request-conditioned skill auditing, benchmark construction and calibrated risk scoring for STARS.

HKUST Academy of Interdisciplinary Studies Feb - May 2026

Remote research intern | Huamin Qu and Yuzhe Shi

Co-led procedural knowledge representation for agent skills, translating narrative procedures into runtime-checkable commitments.

The Hong Kong University of Science and Technology (Guangzhou) Dec 2025 - Jun 2026

Exchange student | Lei Chen

Constructed task-level BrowserAgent SFT data; studied agent memory, state compression and long-context retrieval.

Tsinghua University HCI Lab Dec 2024 - Aug 2025

Remote research assistant | Chun Yu and Yuanchun Shi

Built long-context agentic-RAG benchmarks and retrieval pipelines; contributed memory-aware expert-avatar interaction workflows.

DeepWisdom MetaGPT Oct - Nov 2025

Algorithm evaluation engineer intern

Automated multi-agent coding and MGX benchmarks; used process-level failure analysis to diagnose reliability bottlenecks.

TECHNICAL AND RESEARCH SKILLS

Programming and ML: Python, TypeScript, PyTorch, Hugging Face, PEFT, FastAPI, Docker.
Research methods: controlled experiments, ablation, risk calibration, evaluation design, provenance, reproducible audits, red-teaming.
Agent systems: RAG, tool use, agent memory, multi-model coordination; completed Peking University's AI4MATH course with Lean training.